MEM//VISUALIZER

Dynamic Memory Management Visualiser — Flow Diagrams

Project Documentation • Generated April 2026

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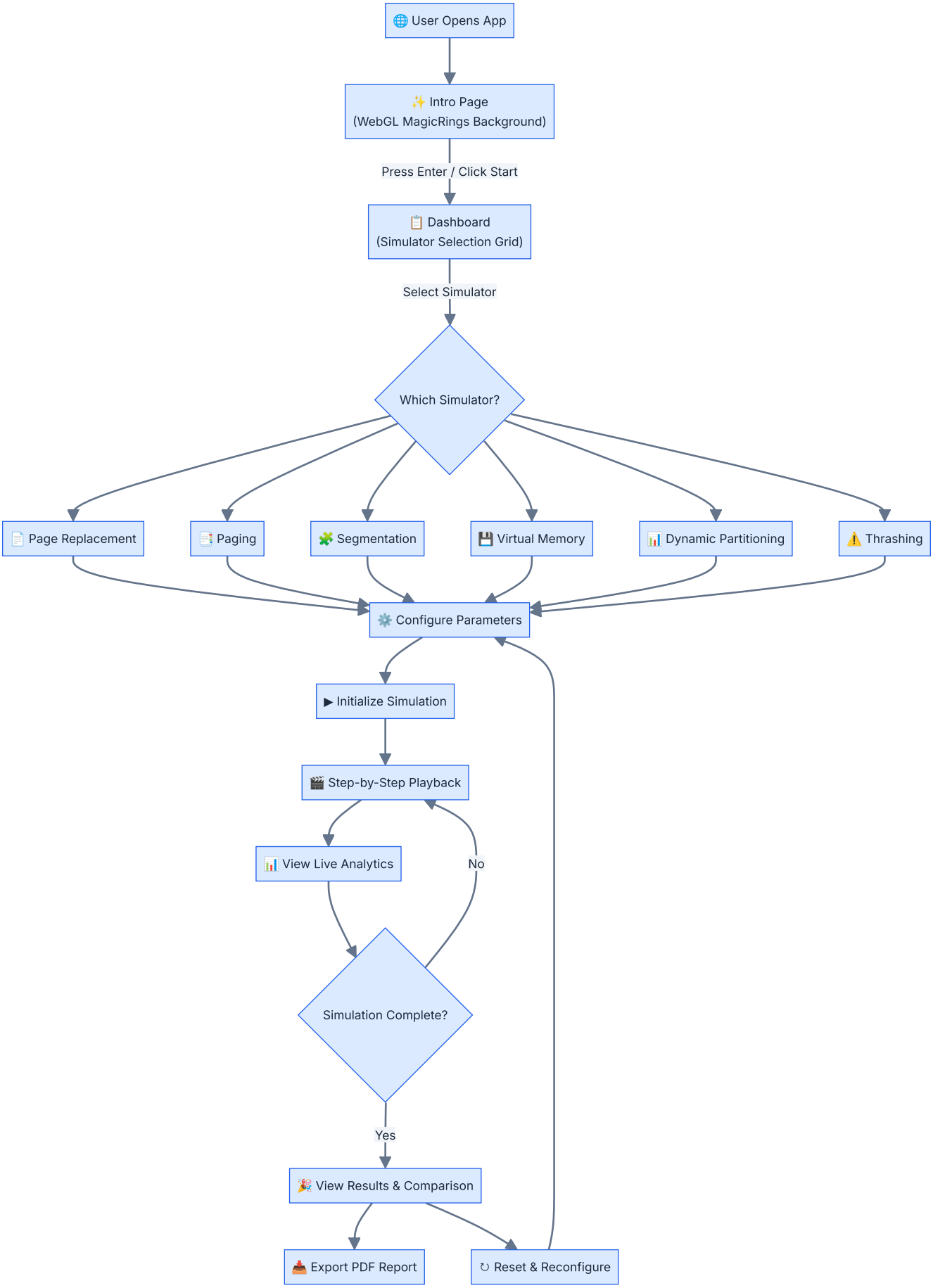
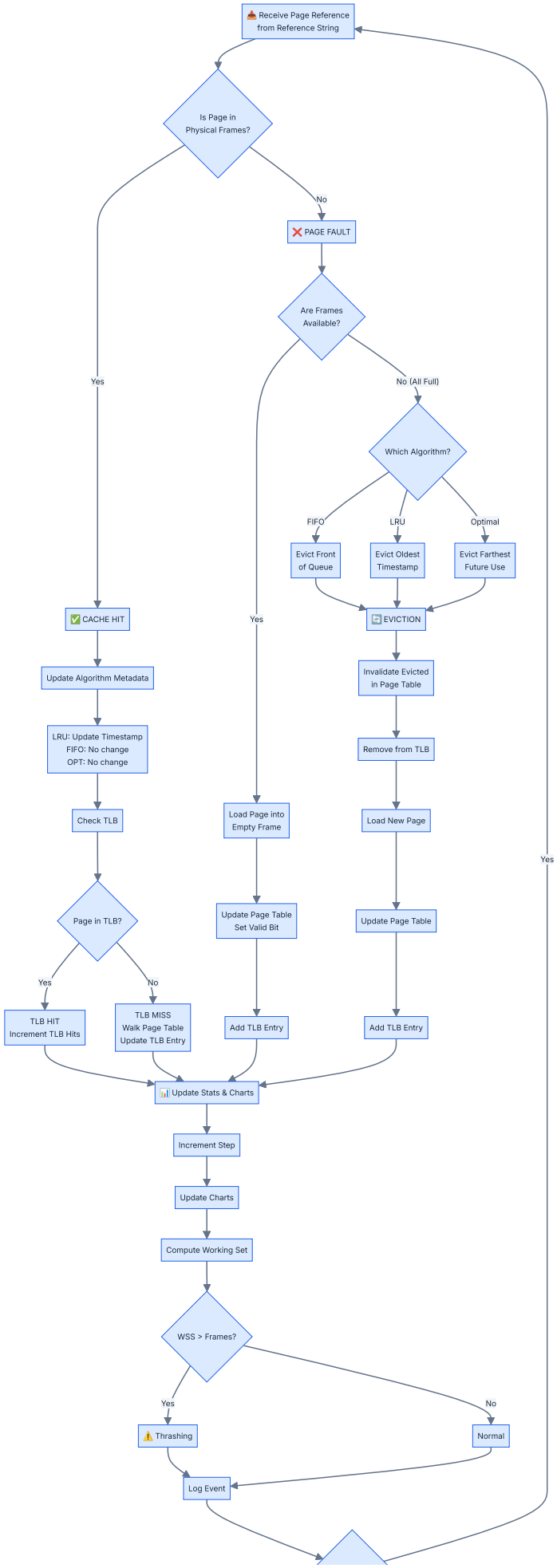
1 Application User Flow

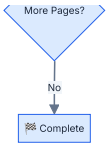
The overall navigation and user journey through the application — from the animated intro landing page, through the dashboard simulator selection grid, into configuration, step-by-step playback, and final results with PDF export.


🌐 User Opens App
✨ Intro Page
(WebGL MagicRings Background)
Press Enter / Click Start
📋 Dashboard
(Simulator Selection Grid)
Select Simulator
Which Simulator?
📄 Page Replacement
📑 Paging
🧩 Segmentation
💾 Virtual Memory
📊 Dynamic Partitioning
⚠️ Thrashing
⚙️ Configure Parameters
▶ Initialize Simulation
🎬 Step-by-Step Playback
📊 View Live Analytics
No
Simulation Complete?
Yes
🎉 View Results & Comparison
📥 Export PDF Report
↻ Reset & Reconfigure

2 Page Replacement Simulation Flow

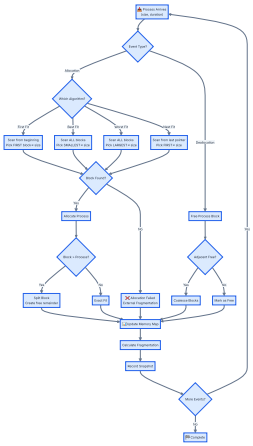
The internal logic when a page reference is processed: checking cache hit/miss, selecting a victim page for eviction based on the chosen algorithm (FIFO/LRU/Optimal), updating the TLB and page table, computing the working set, and detecting thrashing conditions.


📥 Receive Page Reference from Reference String
Is Page in Physical Frames?
Yes
No
❌ PAGE FAULT
Are Frames Available?
Yes
No (All Full)
Which Algorithm?
FIFO
LRU
Optimal
Evict Front of Queue
Evict Oldest Timestamp
Evict Farthest Future Use
✅ CACHE HIT
Yes
🔄 EVICTION
Update Algorithm Metadata
Invalidate Evicted in Page Table
LRU: Update Timestamp
FIFO: No change
OPT: No change
Remove from TLB
Check TLB
Load Page into Empty Frame
Load New Page
Yes
Page in TLB?
Update Page Table Set Valid Bit
Update Page Table
Yes
No
TLB HIT Increment TLB Hits
TLB MISS Walk Page Table Update TLB Entry
Add TLB Entry
Add TLB Entry
📊 Update Stats & Charts
Increment Step
Update Charts
Compute Working Set
WSS > Frames?
Yes
No
⚠️ Thrashing
Normal
Log Event


More Pages?
No
🏁 Complete

3 Dynamic Partitioning Flow

How the partitioning simulator handles memory allocation using four placement algorithms (First Fit, Best Fit, Worst Fit, Next Fit), block splitting when a hole is larger than needed, deallocation with adjacent free block coalescing, and fragmentation metric calculation.

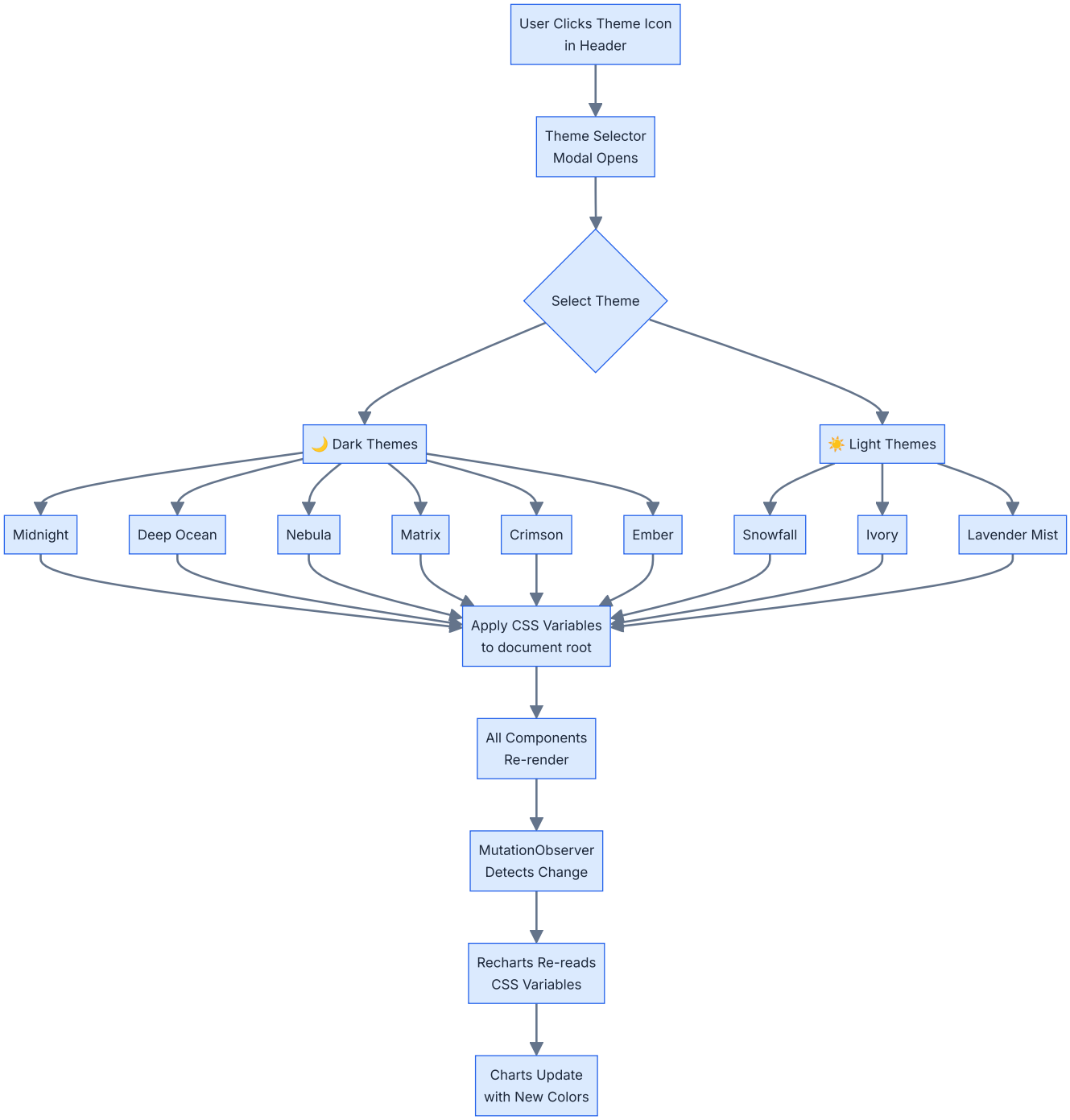

Process Active Data Updated
Event Type?
Allocation
Which algorithm?
FirstFit
Find first beginning big enough gap
BestFit
Scan ALL blocks find SMALLEST fit gap
WorstFit
Scan ALL blocks find LARGEST fit gap
NextFit
Start from last point, resume where left off
Deallocation
Block Found?
No
Allocate Process
Free Process Block
Yes
Yes
Block > Minimum?
Yes
No
Split Block Create free remainder
Exact Fit
Alignment Fail?
Yes
No
Alignment Fail
Coalescing
Mark as Free
Calculate Metrics Map
Calculate Fragmentation
Record Snapshot
More Events?
Yes
Complete

4 Three-Panel UI Architecture

The main simulator interface is organized into three panels: the Config Panel (left) for algorithm selection and playback controls, the Simulation Stage (center) for real-time memory visualization, and the Stats Panel (right) for analytics, charts, and event logging.

5 Theme System Flow

The theme system supports 9 themes (6 dark + 3 light). When a user selects a theme, CSS custom properties are injected into the document root. A MutationObserver detects the change and triggers Recharts to re-read the computed color values for SVG rendering.



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